

Unit IV RNN NLP and Reinforcement Learning

Unit IV – RNN, NLP and Reinforcement Learning

1. Recurrent Neural Network (RNN)

RNN sequential data ke liye use hoti hai jahan current output previous information par depend karta hai. Iska use time-series, speech aur NLP mein hota hai.

Key Feature

RNN hidden state ko maintain karti hai jo previous inputs ko remember kar sakta hai.

2. Vanishing Gradient Problem

RNN training ke dauran gradient bohot chhota ho sakta hai, jisse learning weak ho jati hai. Ye problem vanishing gradient kehlata hai.

3. LSTM

LSTM long short-term memory network hai jo vanishing gradient problem ko solve karta hai. Ye long-term dependencies ko remember kar sakta hai.

Main Components

? Input gate

? Forget gate

? Output gate

4. GRU

GRU gated recurrent unit hai, jo LSTM ka simpler version hai. Isme kam parameters hote hain aur ye fast training provide karta hai.

5. Attention Mechanism

Attention mechanism model ko important words ya positions par focus karne mein help karta hai. Ye sequence models ko improve karta hai.

Use Cases

? Machine translation

? Text summarization

? Image captioning

6. Translation

Translation ka matlab source language ko target language mein convert karna. Sequence-to-sequence models aur transformers isme commonly use hote hain.

7. Beam Search

Beam search best sequence generate karne ka method hai. Ye translation aur text generation mein use hota hai.

How It Works

Multiple candidate sequences ko maintain karke best one select kiya jata hai.

8. BLEU Score

BLEU score machine translation quality evaluate karne ke liye use kiya jata hai. Higher BLEU score generally better quality ko indicate karta hai.

9. Reinforcement Learning Basics

Reinforcement Learning mein agent environment ke saath interact karta hai aur reward ke basis par seekhta hai.

Core Components

? Agent

? Environment

- ? State
- ? Action
- ? Reward

10. RL Framework

Reinforcement learning ka structure action aur reward-driven hota hai. Agent ka goal maximum cumulative reward lana hota hai.

11. MDP

Markov Decision Process ek decision-making framework hai. Ye state transition aur reward model par based hota hai.

Important Terms

- ? State
- ? Action
- ? Transition probability
- ? Reward

12. Bellman Equations

Bellman equations value function ko recursive form mein define karte hain. Ye dynamic programming aur RL mein bahut important hain.

13. Value Iteration and Policy Iteration

Value Iteration

Optimal value function compute karta hai.

Policy Iteration

Policy ko improve karne ka method hai.

14. Q-Learning and SARSA

Q-Learning

Model-free RL algorithm hai aur action-value function seekhta hai.

SARSA

On-policy RL algorithm hai jo current policy ke basis par learn karta hai.

15. Important Exam Points

- ? RNN sequential data ke liye use hoti hai.
- ? LSTM aur GRU vanishing gradient problem ko solve karte hain.
- ? Attention model important words par focus karta hai.
- ? Beam search aur BLEU score NLP mein important hain.
- ? Reinforcement learning mein reward-based learning hoti hai.